DBMS EXPERIMENT NO. 02

ER TO RELATIONAL SCHEMA

AIM

To convert the ER diagram of an Indian e-commerce platform into a relational database schema and implement the schema in MySQL using PRIMARY KEY, FOREIGN KEY, NOT NULL, UNIQUE, ON DELETE CASCADE, and ON DELETE SET NULL constraints. To insert sample data and demonstrate referential integrity violations.

RELATIONAL SCHEMA

- CUSTOMER(CustomerID, Name, Email, DateOfBirth)
- CUSTOMER_PHONE(CustomerID, PhoneNumber)
- ADDRESS(AddressID, Street, City, State, Country, Pincode, AddressType)
- CUSTOMER_ADDRESS(CustomerID, AddressID)
- CATEGORY(CategoryID, CategoryName)
- SELLER(SellerID, SellerName, Email, Phone, GSTIN, AddressID)
- PRODUCT(ProductID, ProductName, Description, Price, StockQuantity, SKU, CategoryID, SellerID)
- PHYSICAL_PRODUCT(ProductID, Weight, Dimensions, ManufacturingDate)
- DIGITAL_PRODUCT(ProductID, FileSize, FileType, DownloadLink)
- ORDERS(OrderID, CustomerID, OrderDate, TotalAmount, OrderStatus)
- ORDER_ITEM(OrderID, ProductID, Quantity, UnitPrice, Discount)
- PAYMENT(PaymentID, OrderID, PaymentDate, Amount, PaymentMethod, PaymentStatus, TransactionID)
- DELIVERY(DeliveryID, OrderID, AddressID, DeliveryDate, DeliveryStatus, CourierPartner, TrackingNumber, EstimatedDeliveryDate, ActualDeliveryDate)

PROGRAM

```
CREATE DATABASE EcommerceDB;
USE EcommerceDB;

CREATE TABLE Customer (
    CustomerID INT PRIMARY KEY,
    Name VARCHAR(100) NOT NULL,
    Email VARCHAR(100) NOT NULL UNIQUE,
    DateOfBirth DATE
);

CREATE TABLE Customer_Phone (
    CustomerID INT,
    PhoneNumber VARCHAR(15),
    PRIMARY KEY (CustomerID, PhoneNumber),
    FOREIGN KEY (CustomerID)
        REFERENCES Customer(CustomerID)
        ON DELETE CASCADE
);

CREATE TABLE Address (
    AddressID INT PRIMARY KEY,
    Street VARCHAR(150) NOT NULL,
    City VARCHAR(50) NOT NULL,
    State VARCHAR(50) NOT NULL,
    Country VARCHAR(50) DEFAULT 'India',
```

```
    Pincode VARCHAR(10) NOT NULL,
    AddressType VARCHAR(30)
);

CREATE TABLE Customer_Address (
    CustomerID INT,
    AddressID INT,
    PRIMARY KEY (CustomerID, AddressID),
    FOREIGN KEY (CustomerID)
        REFERENCES Customer(CustomerID)
        ON DELETE CASCADE,
    FOREIGN KEY (AddressID)
        REFERENCES Address(AddressID)
        ON DELETE CASCADE
);

CREATE TABLE Category (
    CategoryID INT PRIMARY KEY,
    CategoryName VARCHAR(100) NOT NULL UNIQUE
);

CREATE TABLE Seller (
    SellerID INT PRIMARY KEY,
    SellerName VARCHAR(100) NOT NULL,
    Email VARCHAR(100) UNIQUE,
    Phone VARCHAR(15),
    GSTIN VARCHAR(20) UNIQUE,
    AddressID INT,
    FOREIGN KEY (AddressID)
        REFERENCES Address(AddressID)
        ON DELETE SET NULL
);

CREATE TABLE Product (
    ProductID INT PRIMARY KEY,
    ProductName VARCHAR(100) NOT NULL,
    Description VARCHAR(500),
    Price DECIMAL(10,2) NOT NULL,
    StockQuantity INT NOT NULL DEFAULT 0,
    SKU VARCHAR(50) UNIQUE NOT NULL,
    CategoryID INT NOT NULL,
    SellerID INT,
    FOREIGN KEY (CategoryID)
        REFERENCES Category(CategoryID)
        ON DELETE RESTRICT,
    FOREIGN KEY (SellerID)
        REFERENCES Seller(SellerID)
        ON DELETE SET NULL
);

CREATE TABLE Physical_Product (
    ProductID INT PRIMARY KEY,
    Weight DECIMAL(8,2),
    Dimensions VARCHAR(100),
    ManufacturingDate DATE,
    FOREIGN KEY (ProductID)
        REFERENCES Product(ProductID)
        ON DELETE CASCADE
);

CREATE TABLE Digital_Product (
    ProductID INT PRIMARY KEY,
    FileSize DECIMAL(10,2),
```

```
    FileType VARCHAR(50),
    DownloadLink VARCHAR(255),
    FOREIGN KEY (ProductID)
        REFERENCES Product(ProductID)
        ON DELETE CASCADE
);

CREATE TABLE Orders (
    OrderID INT PRIMARY KEY,
    CustomerID INT NOT NULL,
    OrderDate DATE NOT NULL,
    TotalAmount DECIMAL(12,2) NOT NULL,
    OrderStatus VARCHAR(30) DEFAULT 'Pending',
    FOREIGN KEY (CustomerID)
        REFERENCES Customer(CustomerID)
        ON DELETE CASCADE
);

CREATE TABLE Order_Item (
    OrderID INT,
    ProductID INT,
    Quantity INT NOT NULL,
    UnitPrice DECIMAL(10,2) NOT NULL,
    Discount DECIMAL(5,2) DEFAULT 0,
    PRIMARY KEY (OrderID, ProductID),
    FOREIGN KEY (OrderID)
        REFERENCES Orders(OrderID)
        ON DELETE CASCADE,
    FOREIGN KEY (ProductID)
        REFERENCES Product(ProductID)
        ON DELETE CASCADE
);

CREATE TABLE Payment (
    PaymentID INT PRIMARY KEY,
    OrderID INT NOT NULL,
    PaymentDate DATE,
    Amount DECIMAL(12,2) NOT NULL,
    PaymentMethod VARCHAR(30),
    PaymentStatus VARCHAR(30),
    TransactionID VARCHAR(100) UNIQUE,
    FOREIGN KEY (OrderID)
        REFERENCES Orders(OrderID)
        ON DELETE CASCADE
);

CREATE TABLE Delivery (
    DeliveryID INT PRIMARY KEY,
    OrderID INT NOT NULL,
    AddressID INT NOT NULL,
    DeliveryDate DATE,
    DeliveryStatus VARCHAR(30),
    CourierPartner VARCHAR(100),
    TrackingNumber VARCHAR(100) UNIQUE,
    EstimatedDeliveryDate DATE,
    ActualDeliveryDate DATE,
    FOREIGN KEY (OrderID)
        REFERENCES Orders(OrderID)
        ON DELETE CASCADE,
    FOREIGN KEY (AddressID)
        REFERENCES Address(AddressID)
        ON DELETE RESTRICT
);
```

SAMPLE DATA

```
INSERT INTO Customer VALUES
(1, 'Rahul Sharma', 'rahul@gmail.com', '2005-06-15');

INSERT INTO Address VALUES
(101, 'MG Road', 'Lucknow', 'Uttar Pradesh',
 'India', '226001', 'Home');

INSERT INTO Customer_Address VALUES
(1, 101);

INSERT INTO Category VALUES
(1, 'Electronics');

INSERT INTO Seller VALUES
(201, 'Tech Store India', 'seller@gmail.com',
 '9876543210', 'GSTIN12345', 101);

INSERT INTO Product VALUES
(301, 'Laptop', 'AI/ML Laptop', 65000.00,
 10, 'LAP001', 1, 201);

INSERT INTO Physical_Product VALUES
(301, 1.80, '35x24x2 cm', '2026-01-10');

INSERT INTO Orders VALUES
(401, 1, '2026-09-30', 65000.00, 'Pending');

INSERT INTO Order_Item VALUES
(401, 301, 1, 65000.00, 0);

INSERT INTO Payment VALUES
(501, 401, '2026-09-30', 65000.00,
 'UPI', 'Successful', 'TXN123456');

INSERT INTO Delivery VALUES
(601, 401, 101, NULL, 'Processing',
 'BlueDart', 'BD123456', '2026-10-05', NULL);
```

CONSTRAINTS USED

- PRIMARY KEY – uniquely identifies each record.
- FOREIGN KEY – maintains relationships between tables.
- NOT NULL – prevents required fields from containing NULL values.
- UNIQUE – prevents duplicate values in selected columns.
- ON DELETE CASCADE – automatically deletes dependent records when the parent record is deleted.
- ON DELETE SET NULL – sets the foreign-key value to NULL when the referenced parent record is deleted.
- ON DELETE RESTRICT – prevents deletion when dependent records exist.

REFERENTIAL INTEGRITY

The database maintains referential integrity using foreign-key constraints. An attempt to insert a record containing a non-existent foreign-key value results in a referential integrity violation. Similarly, ON DELETE CASCADE, ON DELETE SET NULL, and ON DELETE RESTRICT control what happens when referenced records are deleted.

OUTPUT

- Database EcommerceDB created successfully.
- Tables created: Customer, Customer_Phone, Address, Customer_Address, Category, Seller, Product, Physical_Product, Digital_Product, Orders, Order_Item, Payment, and Delivery.
- Sample records inserted successfully.
- Foreign-key constraints successfully enforce referential integrity.

RESULT

The ER diagram was successfully converted into a relational database schema and implemented in MySQL. The required tables, primary keys, foreign keys, and integrity constraints were created successfully. Sample data was inserted, and referential integrity was maintained using the defined constraints.